

EXHIBIT 19

Exhibit 19: U.S. Patent No. 12,412,924 – EVE 40PL

This claim chart shows how each element of the independent claim of U.S. Patent No. 12,412,924 (“the ’924 Patent”) is met by the EVE 40PL battery cell (“EVE 40PL”), either literally or under the doctrine of equivalents. The materials referenced, and the citations, are merely exemplary and are not exhaustive. LG Energy Solution (“LGES”) reserves the right to rely on different or additional sources and/or different or additional citations to sources that further illustrate infringement. The evidence of infringement provided below is representative in nature and subject to change as discovery commences, including expert discovery. Further, the ALJ has not yet construed certain claim terms of the ’924 Patent. Accordingly, LGES also expressly reserves the right to supplement or amend its claim charts as discovery progresses.

Claims	Exemplary Evidence of Infringement
1[pre]. A separator for a lithium battery comprising:	The EVE 40PL is a lithium battery that includes a separator.  As indicated by EVE Energy North America, the EVE 40PL is a “lithium-ion rechargeable” battery.



Cylindrical NCM
Cells
(18650/21700)



Prismatic LFP Cells

See <https://www.evebatteryusa.com/lithium-ion> (last accessed June 12, 2026).

CYLINDRICAL NCM CELL SPECIFICATIONS



Cell Model	Capacity (Ah)	Nominal Voltage (V)	Max. Cont. Charge/Discharge	Pulse Charge/Discharge	Height (mm)	Diameter (mm)	Weight (g)
40PL(Tabless 21700)	4	3.6	2C/17.5C		70.15	21.15	67

See <https://www.evebatteryusa.com/18650-and-21700-specifications>. (last accessed June 12, 2026).

As shown below, the EVE 40PL includes a separator.

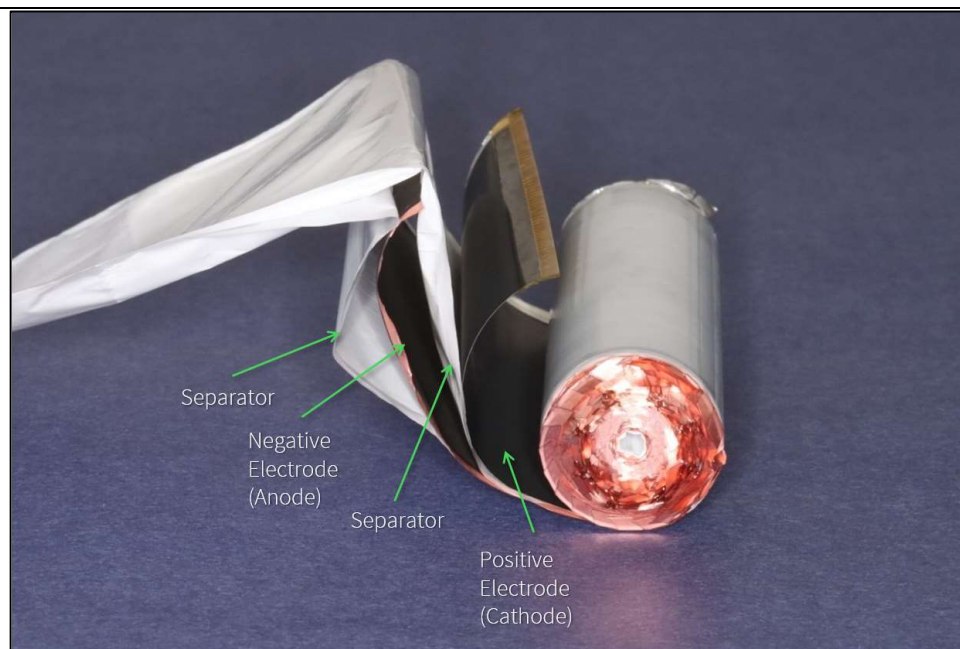
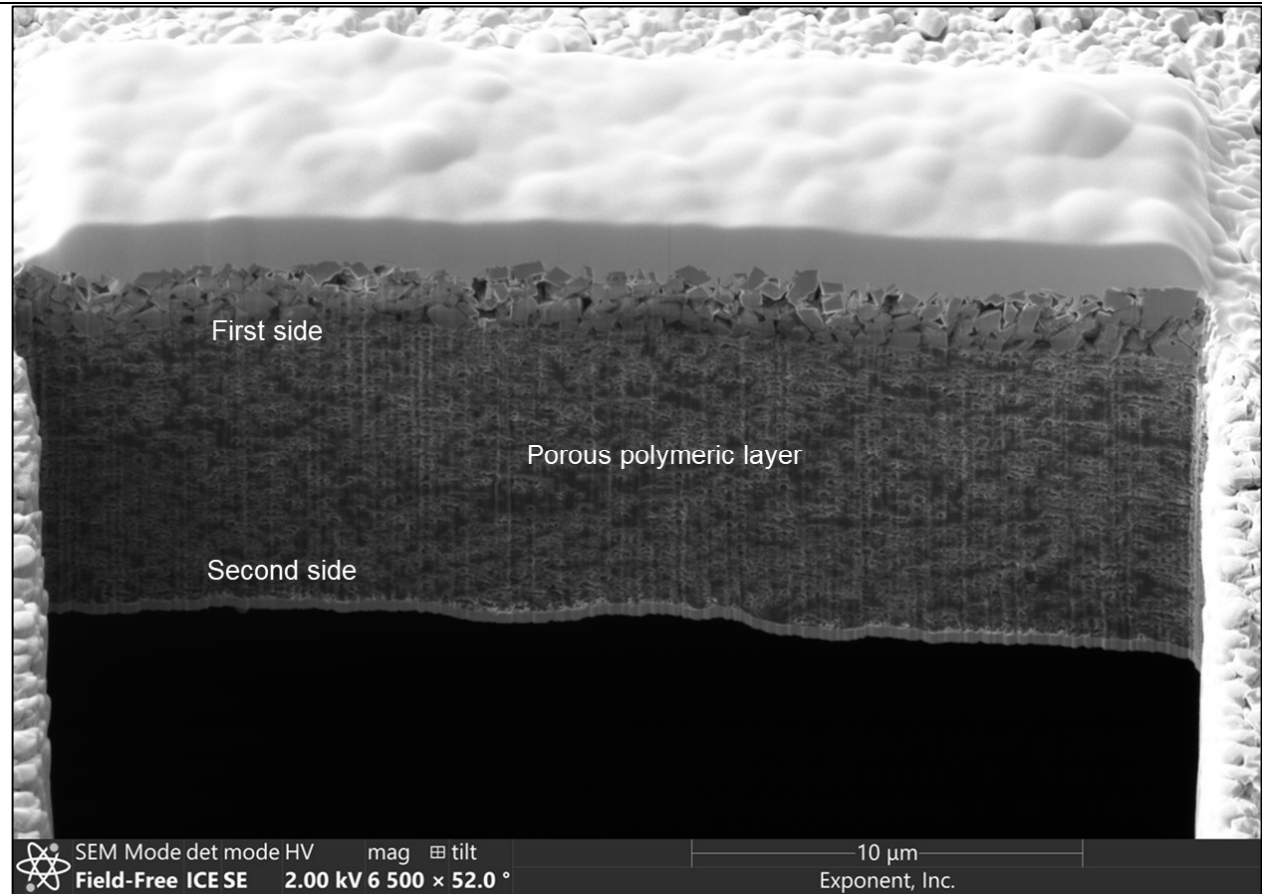


Image of the disassembled electrolyte assembly from the EVE 40PL (annotations added).

1[a]. a porous polymeric layer having a first side and a second side;

The EVE 40PL's separator includes a porous polymeric layer having a first side and a second side.

As shown in the cross-section scanning electron microscopy ("SEM") image below, the separator comprises a porous polymeric layer having a first side and a second side.

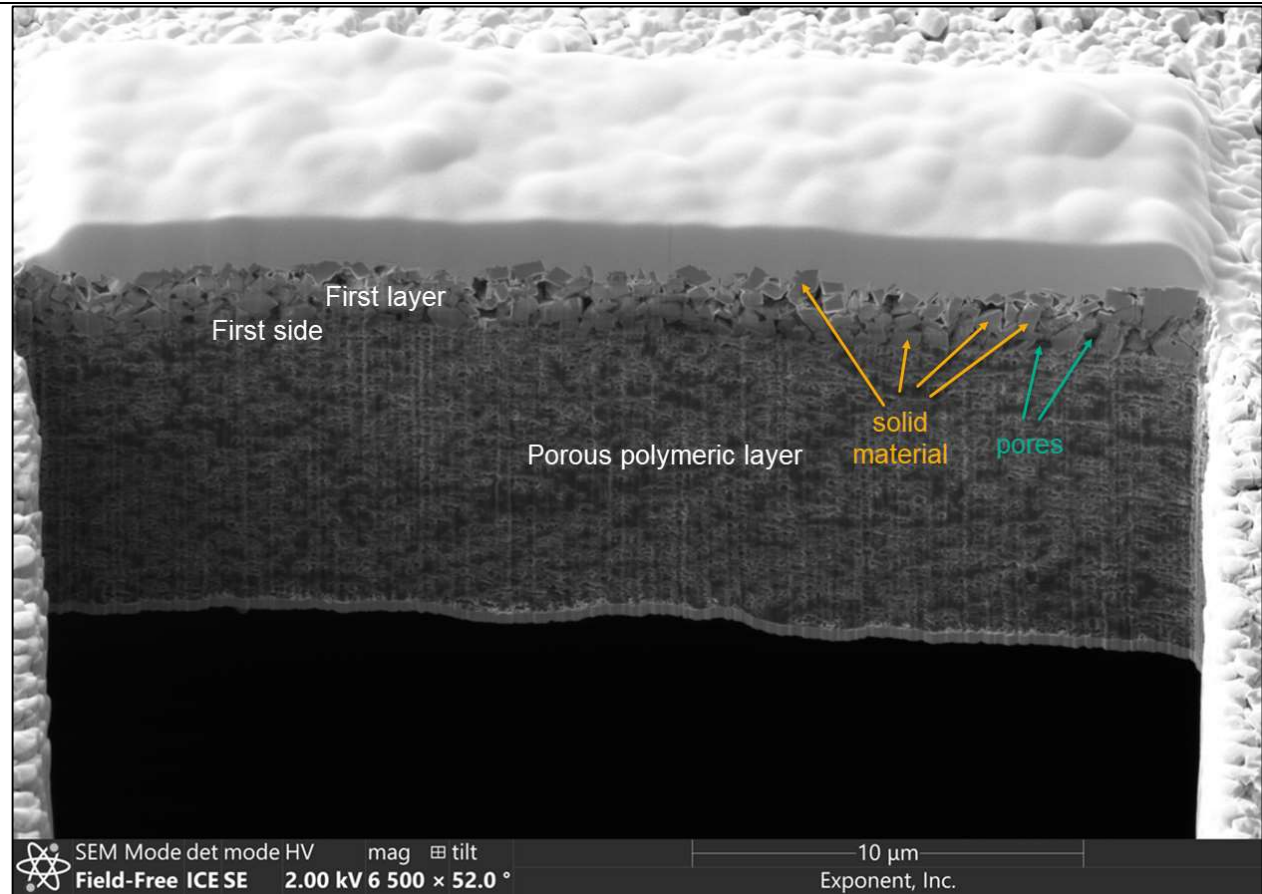


SEM image of the cross section of EVE 40PL separator milled by a focused ion beam (“FIB”) (annotations added).

1[b]. a first layer adjacent the first side of the porous polymeric layer, said first layer comprising pores and solid material, wherein

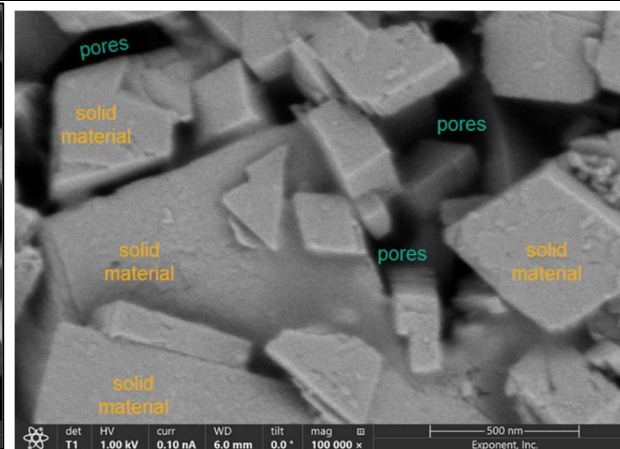
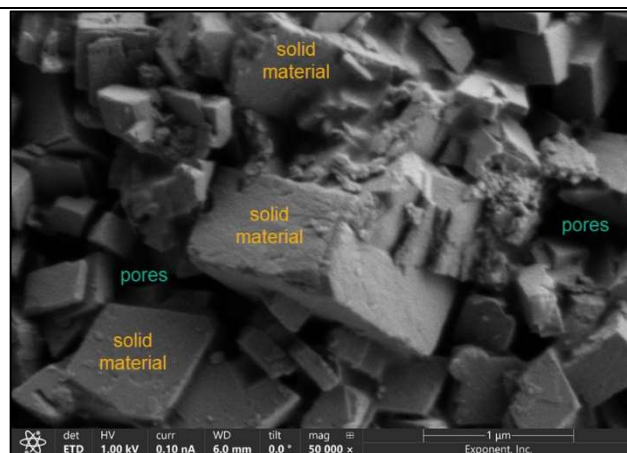
The EVE 40PL’s separator includes a first layer adjacent the first side of the porous polymeric layer, said first layer comprising pores and solid material.

As shown in the cross-section SEM image below, the separator shows a first layer adjacent to the first side of the porous polymeric layer. The image further shows that the first layer includes pores (green) and solid material (orange).



SEM image of the FIB cross-section of EVE 40PL separator (annotations added).

When viewed from top down, the first layer under SEM also shows pores and solid material.

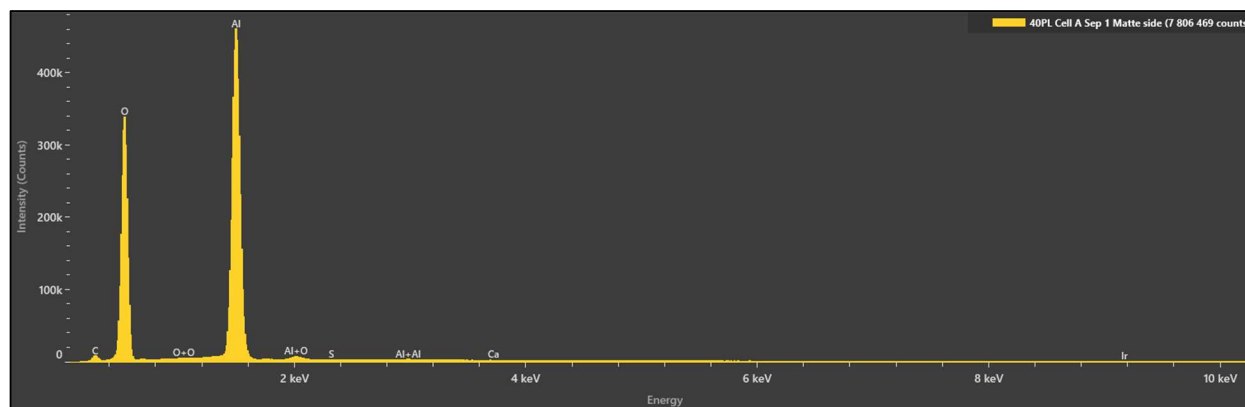


SEM images of the first layer of the EVE 40PL separator (annotations added).

1[c]. (a) the solid material comprises inorganic oxide particles and at least one polymer;

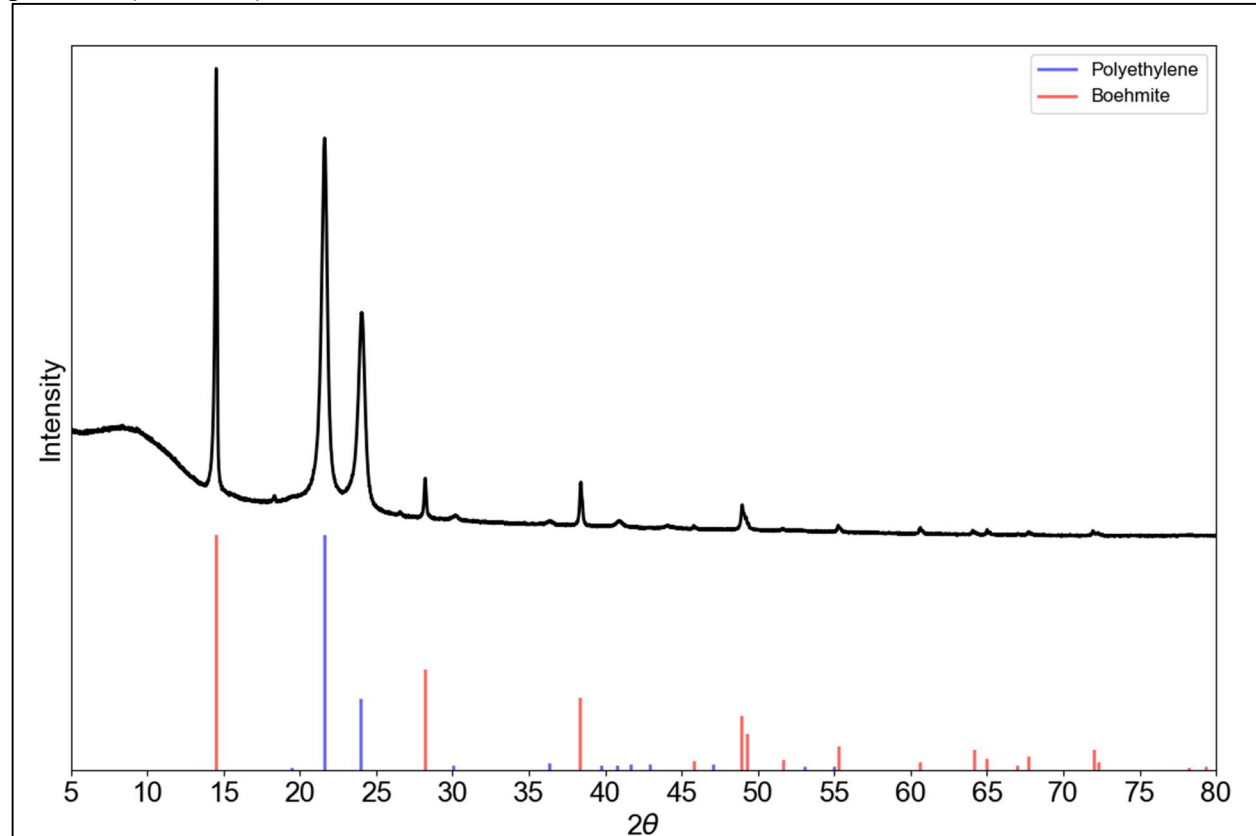
The EVE 40PL's separator includes solid material that comprises inorganic oxide particles and at least one polymer.

Scanning electron microscopy combined with energy dispersive spectroscopy (SEM-EDS) analysis of the solid material confirms the presence of an aluminum oxide compound (boehmite) with strong aluminum and oxygen peaks.



SEM-EDS analysis of the solid material in the first layer of the separator in the EVE 40PL.

X-ray diffraction (XRD) analysis further confirms the solid material comprises inorganic oxide particles (boehmite).



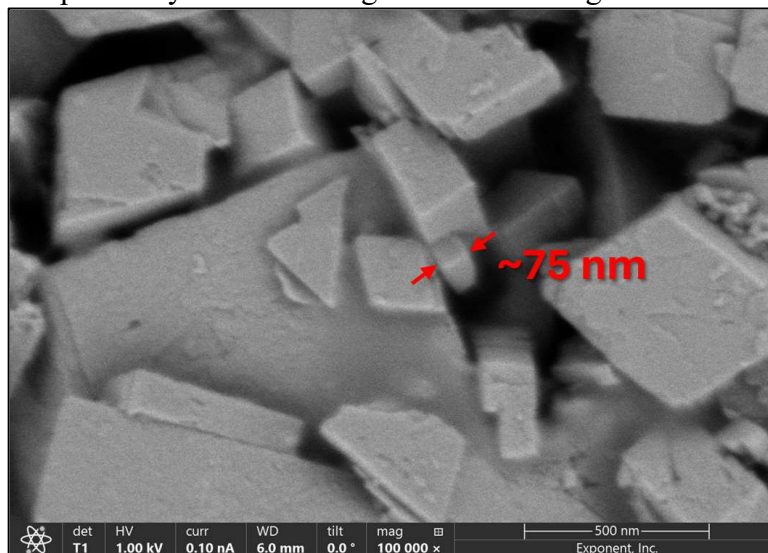
XRD analysis of the solid material in the first layer of the separator in the EVE 40PL.

Further, the boehmite particles adhere to the porous polymeric layer of the separator, as shown in the cross-section and top-down SEM images above. On information and belief, the first layer includes at least one polymer that functions as a binder to allow the boehmite particles to adhere to the surface of the porous polymeric layer.

1[d]. (b) a plurality of the inorganic oxide particles in the first layer comprise crystallites having a size in the range of 5 nm to 90 nm;

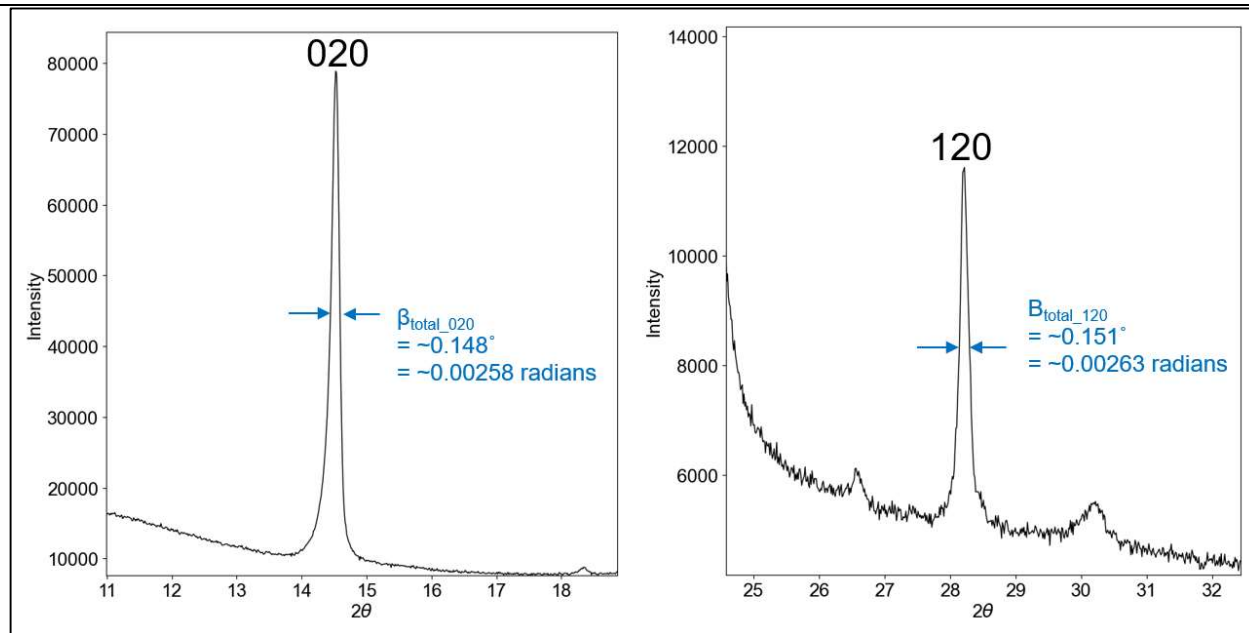
The EVE 40PL's separator includes a plurality of the inorganic oxide particles in the first layer that comprise crystallites having a size in the range of 5 nm to 90 nm.

As shown in the top-down SEM images below, a plurality of the boehmite particles in the first layer comprises crystallites having a size in the range of 5 nm to 90 nm, *e.g.*, ~75 nm.



SEM image of the first layer of the EVE 40PL (annotation added).

Further, using the Scherrer equation, the average crystallite size of the boehmite particles in the first layer is estimated to be ~84 to 85 nm.



XRD analysis for boehmite in the first layer of the EVE 40PL separator.

Scherrer Equation:

$$D = \frac{K\lambda}{\beta_{sample} \cos(\theta)}$$

D = mean crystallite size

K = shape factor, assumed to be 0.9

λ = wavelength of source, 0.154 nm

β_{sample} = corrected full width at half maximum*

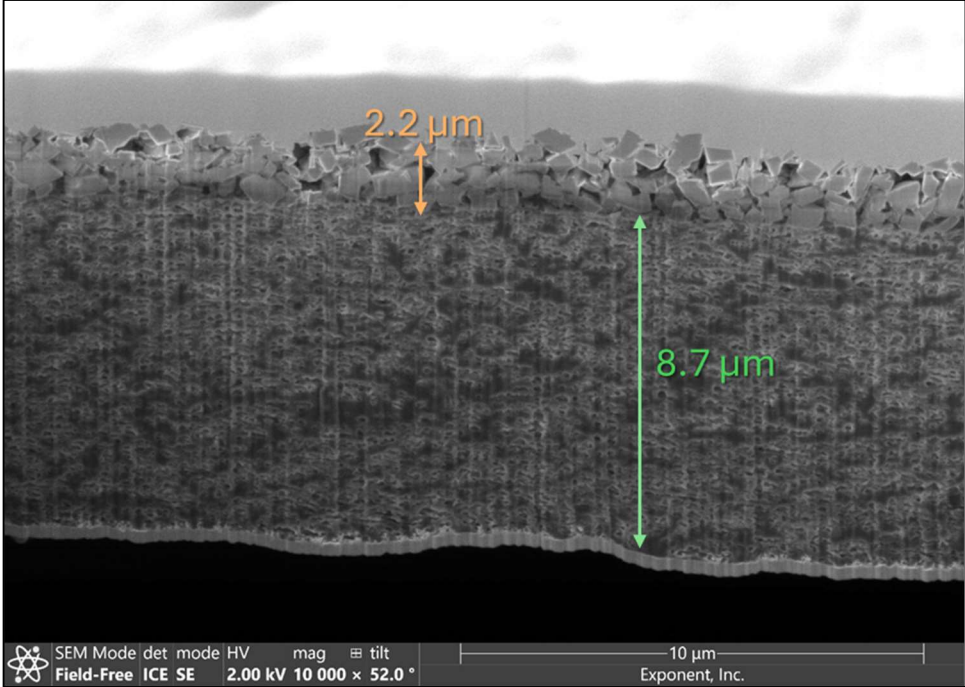
θ = Bragg angle

$$\beta_{total}^{1.5} = \beta_{sample}^{1.5} + \beta_{inst}^{1.5}$$

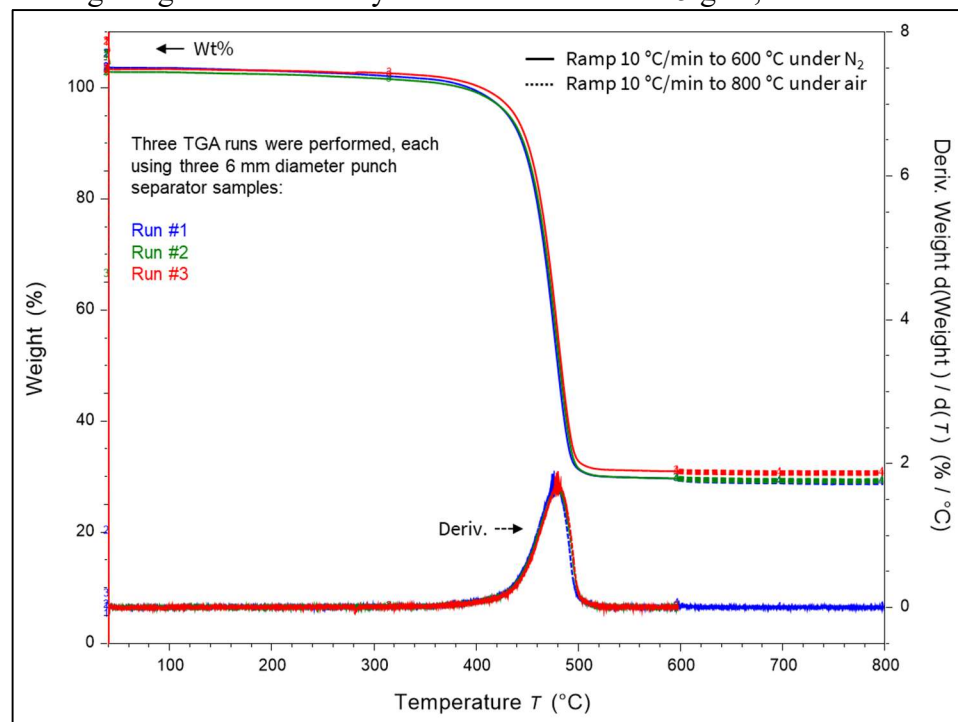
$\beta_{inst} = 0.00161 \text{ rad}$ (same value for both peaks)

$\beta_{sample} = \sim 0.00164 \text{ rad}$ (value for 020 peak)

$\beta_{sample} = \sim 0.00171 \text{ rad}$ (value for 120 peak)

	$D_{020} = \sim 85 \text{ nm}$ (using 020 peak at $2\theta = 14.57^\circ$) $D_{120} = \sim 84 \text{ nm}$ (using 120 peak at $2\theta = 28.31^\circ$) Accordingly, the mean crystallite size of the boehmite was calculated to be $\sim 84\text{-}85 \text{ nm}$ using the XRD results for the 020 and 120 peaks and the Scherrer Equation after correcting for the instrument broadening.
1[e]. (c) a thickness of the first layer is 3 microns or less and a total coating weight of the first layer is less than 6.0 gsm; and	The EVE 40PL's separator has a thickness of the first layer that is 3 microns or less and a total coating weight of the first layer is less than 6.0 gsm. As shown in the cross-section SEM image below, a thickness of the first layer is estimated to be 2.2 microns, which is less than 3 microns.  SEM image of the FIB cross-section of EVE 40PL separator (annotations added).

Based on thermogravimetric analysis (TGA) of the coated separator in the EVE 40PL, a total coating weight of the first layer is estimated to be 2.5 gsm, which is less than 6.0 gsm.



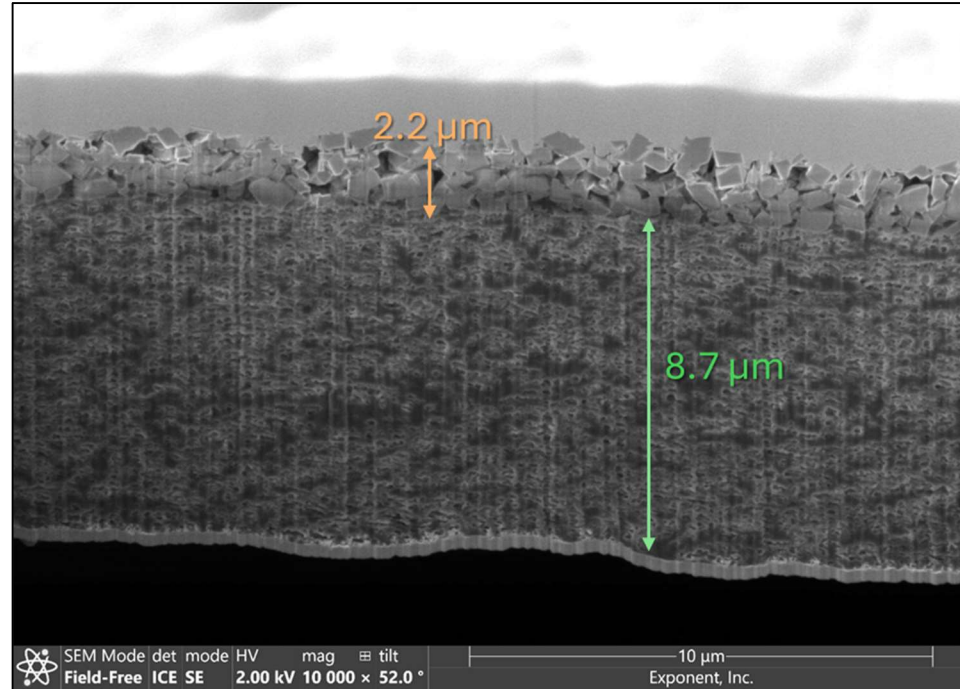
TGA analysis of the separator of the EVE 40PL.

Specifically, the average mass of the 6 mm diameter separator samples was 0.242 mg, which corresponds to a total separator footprint mass of 8.6 g/m². The relative mass ratio of polymeric-to-inorganic material was determined to be 71:29 based on TGA. Therefore, the coating mass was calculated as $8.6 \text{ g/m}^2 \times 0.29 = 2.5 \text{ g/m}^2$, which is less than 6.0 g/m². On information and belief, the presence of binder does not significantly alter this calculation.

1[f]. (d) a ratio of the thickness of the first layer to a total thickness of the separator is 10% to 80%.

The EVE 40PL's separator has a ratio of the thickness of the first layer to a total thickness of the separator between 10% and 80%.

As shown below, a ratio of the thickness of the first layer (approximately 2.2 microns) to the total thickness of the separator (10.9 microns) is approximately 20.2%, which is between 10% and 80%.



SEM image of the FIB cross-section of EVE 40PL separator (annotations added).